MAT 221 - CALCULUS I
Lesson 2-3 - Integration of Rational Functions

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Procedure for Partial Fractions Decomposition

Step 1. If degree of $p(x) \geqslant$ degree of $q(x)$, first divide to obtain

$$\frac{p(x)}{q(x)} = s(x) + \frac{t(x)}{q(x)} \tag{1}$$

where s, t and q are polynomials and degree of $t(x) <$degree of $q(x)$.

Step 2. Factor $q(x)$ into linear and/or irreducible quadratic factors. A linear factor will have the form $x - a_i$, where a_i is a root of $q(x)$, and a quadratic factor will have the form $x^2 + bx + c$.

Step 3. For each factor of the form $(x - a_i)^k$, the partial-fraction decomposition (2) contains a sum of k partial fractions.

$$\frac{A_1}{x - a_i} + \frac{A_2}{(x - a_i)^2} + \ldots + \frac{A_k}{(x - a_i)^k}. \tag{2}$$

Step 4. For each factor of the form $(x^2 + bx + c)^m$, the partial-fraction decomposition contains a sum of m partial fractions

$$\frac{A_1 x + B_1}{x^2 + bx + c} + \frac{A_2 x + B_2}{(x^2 + bx + c)^2} + \ldots + \frac{A_m x + B_m}{(x^2 + bx + c)^m}. \tag{3}$$

Example 1

Compute

$(a)\ \int \frac{dx}{x^2+4x-5}$ $(b)\ \int \frac{dx}{x(x-1)^2}$

$(c)\ \int \frac{x^3-x^2+4x-5}{x-2}dx$ $(d)\ \int \frac{dx}{x^3+x}$

$(e)\ \int_0^{\pi/2} \frac{\cos x}{\sin^2 x+2\sin x+2}dx$ $(f)\ \int_0^1 \frac{dx}{(x-2)^2(x-3)^2}$

Solution

(a) First factor the denominator; $x^2+4x-5=(x+5)(x-1)$. Then we have

$$\frac{1}{x^2+4x-5}=\frac{1}{(x+5)(x-1)}=\frac{A}{x+5}+\frac{B}{x-1}.$$

Using cover up rule we get $A=-\frac{1}{6}$ and $B=\frac{1}{6}$.

Thus, $\frac{1}{x^2+4x-5}=\frac{-\frac{1}{6}}{x+5}+\frac{\frac{1}{6}}{x-1}$ and

$$\int \frac{dx}{x^2+4x-5}=\frac{1}{6}\int\left(\frac{1}{x-1}-\frac{1}{x+5}\right)dx=\frac{1}{6}[\ln|x-1|-\ln|x+5|]$$

$$=\frac{1}{6}\ln\left|\frac{x-1}{x+5}\right|+C$$

(b)-(f) in class

Integrals of the form: $\int \frac{dz}{a^2 - z^2}$ and $\int \frac{dz}{a^2 + z^2}$

By completing squares of the denominator and using other techniques of integration it can be shown that

$$\int \frac{dz}{a^2 - z^2} = \frac{1}{2a} \ln \left| \frac{a + z}{a - z} \right| + C \tag{4}$$

$$\int \frac{dz}{a^2 + z^2} = \frac{1}{a} \tan^{-1} \left(\frac{z}{a} \right) + C \tag{5}$$

Example 2 Integrate with respect to x.

$(a) \int \frac{dx}{9 - x^2}$ $(b) \int \frac{dx}{x^2 + 16}$ $(c) \int \frac{dx}{5 + 4x - 2x^2}$

$(d) \int \frac{dx}{x^2 - 2x + 5}$ $(e) \int \frac{dx}{x^2 + 4x - 5} dx$ $(f) \int \frac{x + 8}{x^2 + 6x + 12} dx$

solution

(a) $\int \frac{dx}{9 - x^2} = \int \frac{dx}{3^2 - x^2} = \frac{1}{2(3)} \ln \left| \frac{3 + x}{3 - x} \right| = \frac{1}{6} \ln \left| \frac{3 + x}{3 - x} \right| + C$

(b) - (f) solution to be done in class.